Abel Thomas Babu

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Newcastle Upon Tyne

SUMMARY

Software Engineer specialising in Python backend systems and applied AI, with experience building APIs, automation tools, and performance-optimised ML pipelines. Currently building toward AI engineering roles centred on LLMs and production RAG systems.

EDUCATION

Newcastle University, MSc. Advanced Computer Science (Distinction) *Newcastle Upon Tyne, UK*

Sept 2023 – Dec 2024

Relevant Modules: Engineering for AI, Machine Learning, Cloud Computing, Enterprise Middleware

- Dissertation: built a vision-driven reinforcement learning agent merging a YOLOv8s object-detection model with a Deep Q-Network (Stable-Baselines3) in a custom Gymnasium environment; rebuilt the capture pipeline to cut inference latency from 64–67ms to 9–14.5ms and reached 0.994 mAP@0.5, generalising across three different games with no retraining.
- Awarded the Vice Chancellor's International Scholarship following a written application setting out my case for the award.
- Received the NCL+ Award for extracurricular development, recognising volunteering (British Heart Foundation) alongside self-directed learning outside the core curriculum.

Saintgits College, B.Tech in Computer Science and Engineering (First Class) *Kerala, India*

Apr 2018 – Apr 2022

Relevant Modules: Data mining and warehousing, Data Structures, OOP

- Final-year project: co-developed an IoT-based flood prediction and early-warning system as part of a 4-person team; integrated the hardware layer (Arduino UNO, ESP8266 Wi-Fi module, DHT22 temperature/humidity sensor, HC-SR04 ultrasonic water-level sensor) feeding a real-time Firebase database, and built the visualisation of the logistic regression model's flood/no-flood classification.

RELEVANT EXPERIENCE

Nezuna Technologies Inc. | *C#/.NET 9, PostgreSQL, React + TypeScript, Flutter* *Aug 2026 – Present*

Software Engineer *Remote, Toronto*

- **Drag-and-drop shift scheduler:** built it end to end across Month, Week and Day views (React, dnd-kit), with draft-based editing, undo/redo, multi-select and copy/cut/paste across months, backed by a new .NET endpoint that saves a whole batch in one database transaction, rejects double-bookings and notifies guards once rather than per change.
- **Client invoicing module:** built the .NET backend for site bill rates with history, invoice generation from attendance records, gap-free invoice numbering, send/resend/void and PDF rendering; shipped it in the office web app and the Flutter client app, which refreshes on push notifications.
- **Incident report PDFs:** generated on demand rather than stored, with photos rotated and downscaled correctly and times shown in the account's time zone.

Apps Team Technologies | *Swift, Xcode, UIKit* *Dec 2022 – Feb 2023*

Software Developer Intern *Kerala, India*

- Designed and implemented core UI screens and reusable components for a logistics iOS application, improving interaction speed and reducing navigation latency.
- Resolved a list-rendering inefficiency that improved scrolling performance by approximately 35% across major UI modules.
- Integrated backend RESTful APIs with Swift asynchronous operations, ensuring seamless live data sync, reliable error handling, and responsive frontend UI updates.
- Facilitated a newly joined intern on Swift and UIKit basics, compiling structured Notion notes and codebase documentation to accelerate transition.
- Collaborated cross-functionally with engineers, designers, and stakeholders to refine application workflows and ensure accurate integration between frontend interfaces and backend APIs.

PROJECTS

AI Papers RAG Assistant | Python, LangChain, OpenAI API, ChromaDB, Docker, Kubernetes

- Built a retrieval-augmented generation (RAG) chatbot answering questions over a custom document, combining dense vector search with BM25 keyword retrieval via LangChain's EnsembleRetriever for hybrid retrieval.
- Achieved 0.94 faithfulness and 0.94 context recall (RAGAS evaluation framework) across a held-out test set, validating retrieval accuracy and generation groundedness.
- Engineered a chunking and embedding pipeline (OpenAI text-embedding-3-small) with persistent vector storage in ChromaDB and enforced source citations on every generated answer.
- Containerised the application with Docker (multi-stage build, health checks, non-root user) and deployed to a local Kubernetes cluster using Deployments, Services, ConfigMaps, Secrets, and PersistentVolumeClaims.
- Shipped an interactive Streamlit chat interface exposing retrieved source chunks alongside each answer for traceability.

Vision-Based AI for Idle Game Automation | *Python, OpenCV, PyTorch, YOLOv8, Gymnasium, Reinforcement Learning*

- Built a vision-driven automation agent integrating a YOLOv8s object detection model with a Deep Q-Network (DQN) agent (Stable-Baselines3) in a custom Gymnasium environment, without any game-specific logic.
- Re-architected the capture pipeline (multi-threaded, MSS over PyAutoGUI) to cut inference time from 64–67ms to 9–14.5ms and raise frame rate from under 20 FPS to a stable 30 FPS, enabling real-time deployment on consumer-grade hardware (Ryzen 7 6800H, RTX 3050 4GB) at under 70% GPU utilisation.
- Trained the DQN agent over 990,000+ steps using a shaped, multi-component reward function (score progression, action-specific bonuses, decaying exploration bonus), redesigning and expanding the replay buffer to 100,000 after early runs revealed unwanted action bias.
- Unified model reached 0.994 mAP@0.5 and generalised across three different games with no retraining.

Full-Stack Contact Manager | *Python, React, Flask, SQLAlchemy*

- Created a full-stack CRUD contact management application with a responsive React interface.
- Developed RESTful API endpoints supporting create, read, update, delete, and search operations.
- Architected relational database models with SQLAlchemy and optimised query performance.
- Added reusable components and state management with React hooks.

SKILLS

- **Programming Languages:** Python,JavaScript, Java, C#, Swift, SQL
- **Backend / APIs:** PostgreSQL, FastAPI, Flask, REST APIs
- **Frontend / Mobile:** React, Tailwind CSS, Flutter
- **AI / ML:** PyTorch, OpenCV, scikit-learn, LangChain, AI-assisted development (Claude Code)
- **Tools & Platforms:** Git, GitHub Actions, Docker, Kubernetes